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Proposal: Musical Prosody using MIDI

I find myself drawn to the effect that non-lexical factors, like cadence and volume, have on the meaning of spoken language audio data. I propose a project that will apply some conventional audio engineering to format a discrete collection of words (I may explore forced alignment as a different project) as a MIDI file and then classifying that MIDI data as emotional subtext of the original audio. My goal will be to train an LSTM on transcribed MIDI data (the same word labels from the original transcripts, but with MIDI information used to describe the temporal properties) labeled with an emotional subtext to predict those emotional cues from a given MIDI file using the musical features that are exposed in a MIDI file like volume, tempo and rhythm and pitch/melody. The ESD from HLTsingapore looks like a good set of labels – but I will need to tokenize it first with forced alignment (perhaps with whisperX or Montreal Forced Aligner).

How does constraining the information to a representation via MIDI distort the model's ability to make predictions? Is a MIDI file (which is much smaller than a high-quality audio file) a viable alternative to an audio file when predicting emotion?

Plenty of others have researched prosody and its effect on meaning. When previous classification was based on utterance-level averages (the mean of angry and happy is neutral [#2 below]) then flat and neutral outcomes where the most prevalent. Another historical issue has been multiple/disagreeing targets (there's no single way to sound "angry" [#1 below]) which occurred less when researchers used more well-labeled data. I hypothesize that the model trained on MIDI will fail to classify longer samples or more ambiguous samples, but still perform above a baseline of random guessing.

As motivation, I wonder if looking at the prosody as musical notation could change how we think about prosody, meaning, and NLP. I'm curious about what sort of musical patterns might become visible that may have been previously hidden inside the networks as states or weight matrices.

Citations

1. Voice Synthesis Improvement by Machine Learning of Natural Prosody
2. https://doi.org/10.48550/arxiv.2110.02952
3. https://github.com/m-bain/whisperx
4. https://montreal-forced-aligner.readthedocs.io/en/latest/
5. https://github.com/philipperemy/timit
6. https://github.com/Helsinki-NLP/prosody
7. https://github.com/HLTSingapore/Emotional-Speech-Data